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THE HISTORY OF

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THE DYNABOOK

A personal computer for children of all ages: the 1968-1972 vision that personal computing keeps almost fulfilling

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THE CARDBOARD MODEL

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The Dynabook is the most influential computer never built. Alan Kay conceived it between 1968 and 1972: a dynamic medium the size of a notebook, under two pounds, with a keyboard and stylus and network connection, owned by a child. Kay carried a cardboard mockup around Xerox PARC to keep the target concrete. The canonical statement is his 1972 paper "A Personal Computer for Children of All Ages" [1], which opens with a scene of two children playing a physics game on their personal machines in a park -- and then insists the machine's real purpose is not consumption but authorship: simulating, composing, and reasoning in a medium as fluid as paper and as alive as a conversation.

The intellectual scaffolding came from Seymour Papert's work with Logo and children at MIT (later crystallized in Mindstorms [2]), from Jerome Bruner's theories of enactive, iconic, and symbolic learning, and from Marshall McLuhan: the medium is the message, so build the medium that makes every reader a writer.

THE INTERIM DYNABOOK

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Because the real Dynabook was beyond 1970s hardware, PARC built "interim Dynabooks": the Alto workstation running Smalltalk (see the Smalltalk chapter). Bitmapped display, overlapping windows, children scripting animations -- the Alto/Smalltalk system was explicitly a desk-bound stand-in for the notebook in the park. Kay's 1977 Scientific American article with Adele Goldberg, "Personal Dynamic Media," restated the vision as the metamedium: a single device that can simulate all other media.

Douglas Engelbart's 1968 "Mother of All Demos" [3] is the sibling vision -- augmentation of adult knowledge workers where Kay aimed at children's learning -- and the two programs together define the axis along which personal computing has oscillated ever since: empowering tool versus consumption appliance.

ALMOST-DYNABOOKS

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The lineage of hardware that looks like the Dynabook is long: the GRiD Compass, the Apple Newton and the 1987 Knowledge Navigator concept film (see the Hypertext chapter), the laptop itself, the OLPC XO-1 -- which shipped Squeak Etoys and is the closest institutional attempt at the actual program -- and the iPad, which Kay pointedly declined to bless: the form factor arrived without the authorship. A tablet on which a child cannot inspect, modify, and build the software she uses is, in Kay's terms, television with a glass keyboard.

That critique keeps the Dynabook alive as a measuring stick. Kay's own later efforts (Squeak, Etoys, the Viewpoints STEPS program [4]) and the systems in this series' final chapter -- Dynamicland above all -- are explicit attempts to finish the

Dynabook program: not a device but a literacy, in which simulation and dynamic media are things ordinary people read and write. The revolution the cardboard model promised is, by its inventor's own accounting, still pending.

SOURCES & CITATIONS

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- [3] The Mother of All Demos, presented by Douglas Engelbart (1968), on YouTube
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- [4] Viewpoints Research Institute writings: Kay's Dynabook papers and the STEPS reports
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- [5] Alan Kay, Doing with Images Makes Symbols (1987) -- the Dynabook/UI lineage told by its author, Internet Archive
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